

Mask-Guided Cotton Boll Phenotyping Architecture

Pre/post-defoliation UAV imagery → detector-prompted lint masks → morphology 2.5D review → conservative trait summaries

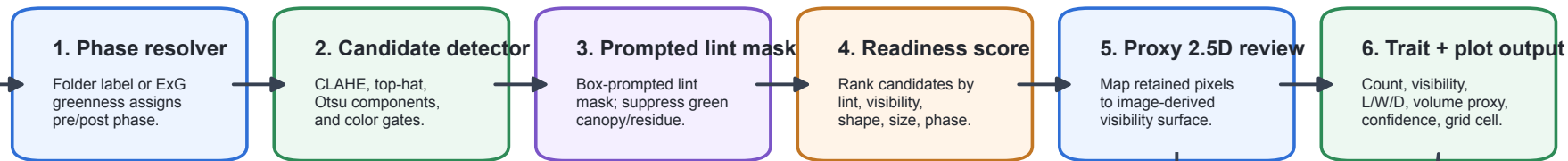
Inputs: aerial visibility intervention
Pre-defoliation RGB



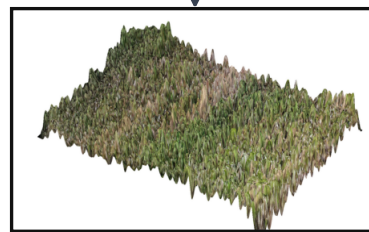
Post-defoliation RGB



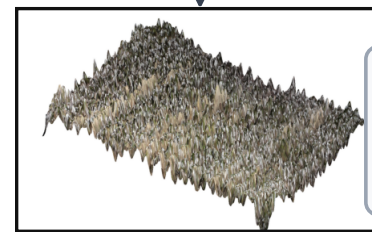
Implemented proxy pipeline



Evidence and outputs



Pre-defoliation proxy 2.5D



Post-defoliation proxy 2.5D

Structured phenotyping record

Candidate table: mask, visibility, axes, diameter proxy, volume proxy, readiness, phase, plot cell.

Calibration-dependent branch

Optional calibrated geometry

Only used when scale, camera intrinsics, pose, GCP/RTK, RGB-D, SfM/MVS, DUST3R/MASt3R/VGGT, or validated 3DGS evidence is available.

Metric trait validation

Report trait MAE/RMSE, mask IoU, reprojection consistency, Chamfer distance, and plot-cell agreement.

Arrow convention

—→ implemented proxy path
- - -→ calibration-only validation

Scientific boundary: 2.5D maps are image-derived aerial visibility surfaces; metric boll geometry requires calibrated scale/camera validation.